BE-102 · BTech First Year · UIT, Barkatullah University, Bhopal

Engineering Chemistry — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Industrial Water and Treatment

1. Sources, Impurities and Hardness

Sources: surface water (rain – sabse pure; river, lake; sea – sabse impure, ~3.5% salts) aur underground water (springs, wells – clear par dissolved salts zyada). **Impurities: dissolved** (Ca/Mg ke HCO_3^-, Cl^-, SO_4^{2-}; gases CO_2, O_2), **suspended** (sand, clay – turbidity), **colloidal** (1–100 nm, clay, $Fe(OH)_3$), **biological** (bacteria, algae).

HARDNESS Water ki woh property jisme soap lather nahi banata, kyunki Ca^{2+}/Mg^{2+} soap ko insoluble scum bana dete hain: $2C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2NaCl$. **Temporary** (carbonate): Ca/Mg bicarbonates – boiling se hatti hai: $Ca(HCO_3)_2 \rightarrow CaCO_3\downarrow + H_2O + CO_2$; $Mg(HCO_3)_2 \rightarrow Mg(OH)_2\downarrow + 2CO_2$. **Permanent** (non-carbonate): Ca/Mg chlorides & sulphates – boiling se nahi jaati.

FORMULA

$$\text{Hardness as } CaCO_3 \text{ eq.} = \frac{\text{Mass of salt (mg/L)} \times 100}{\text{Mol. wt of salt}} \quad (\text{bivalent salts; } CaCO_3\ M = 100, \text{eq. wt } 50)$$

Mol. wt: $Ca(HCO_3)_2$ 162, $Mg(HCO_3)_2$ 146, $CaSO_4$ 136, $CaCl_2$ 111, $MgSO_4$ 120, $MgCl_2$ 95, $MgCO_3$ 84, $Mg(NO_3)_2$ 148. **Units:** 1 ppm = 1 mg/L = 0.1 °Fr = 0.07 °Cl. NaCl, KCl, Na_2SO_4 hardness nahi dete.

2. Water Quality and Analysis

Drinking water (BIS): pH 6.5–8.5, TDS < 500 mg/L, hardness < 200 mg/L, Cl^- < 250 mg/L, pathogen-free. **Boiler feed:** hardness ≈ 0, no dissolved O_2/CO_2. **Chloride** – Mohr's method ($AgNO_3$, K_2CrO_4 indicator, brick-red end point). **DO** – Winkler's method ($MnSO_4$ + alkaline KI, I_2 vs $Na_2S_2O_3$).

Hardness by EDTA method

Buffer $NH_4Cl + NH_4OH$ (pH 10), indicator **Eriochrome Black-T**.

M^{2+} + EBT → M–EBT (**wine red**); M–EBT + EDTA → M–EDTA + EBT (**blue**). Titrate 50 mL each: standard hard water (1 mg $CaCO_3$/mL) = V_1, sample = V_2, boiled sample = V_3.

Total = $(V_2/V_1)\times1000$ ppm; Permanent = $(V_3/V_1)\times1000$ ppm; Temp = Total − Perm.

Alkalinity (vs std. HCl)

P = phenolphthalein end pt ($OH^- + \frac{1}{2}CO_3^{2-}$), **M** = methyl orange (total). OH^- & HCO_3^- saath nahi rehte.

	OH^-	CO_3^{2-}	HCO_3^-
P = 0	0	0	M
P = ½M	0	2P	0
P < ½M	0	2P	M−2P
P > ½M	2P−M	2(M−P)	0
P = M	M	0	0

3. Sterilization of Water

Boiling (small scale). **Chlorination:** $Cl_2 + H_2O \rightarrow HOCl + HCl$; $HOCl \rightarrow HCl + [O]$ – HOCl bacteria ke enzymes destroy karta hai. **Bleaching powder:** $CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$. **Chloramine** ($Cl_2 + NH_3 \rightarrow NH_2Cl + HCl$) – long lasting. **Break-point chlorination:** itna Cl_2 ki organic matter, NH_3, bacteria sab oxidise ho jayein; uske baad free residual Cl_2 badhta hai. **Ozonation:** $O_3 \rightarrow O_2 + [O]$ (costly, no residual). **UV** treatment.

4. Water Softening Processes

(a) Lime–Soda Process

Lime $Ca(OH)_2$ + soda Na_2CO_3 soluble salts ko $CaCO_3\downarrow$ aur $Mg(OH)_2\downarrow$ mein badalte hain. **Cold L-S**: room temp, coagulant ($NaAlO_2$, alum) chahiye, residual 50–60 ppm. **Hot L-S**: 80–150 °C, fast, residual 15–30 ppm.

$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$; $Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3 + Mg(OH)_2 + 2H_2O$; $MgCl_2 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaCl_2$; $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl$; $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$; $CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$.

FORMULA (all in $CaCO_3$ eq. mg/L; × Vol (L) × 100/%purity)

$$\text{Lime} = \frac{74}{100}[\text{Temp } Ca^{2+} + 2 \times \text{Temp } Mg^{2+} + \text{Perm } (Mg^{2+}, Fe^{2+}, Al^{3+}) + CO_2 + H^+ (HCl/H_2SO_4) + HCO_3^- (NaHCO_3)]$$

$$\text{Soda} = \frac{106}{100}[\text{Perm } (Ca^{2+}, Mg^{2+}, Fe^{2+}, Al^{3+}) + H^+ (HCl/H_2SO_4) - HCO_3^- (NaHCO_3)]$$

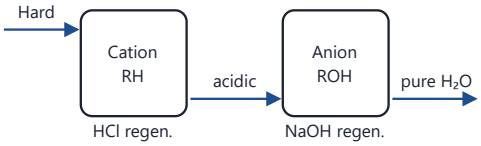
Yaad rakho: Mg permanent hardness ko lime aur soda dono chahiye ($MgCl_2 \rightarrow CaCl_2 \rightarrow$ soda).

(b) Zeolite (Permutit) Process

Zeolite = hydrated sodium aluminosilicate $Na_2O{\cdot}Al_2O_3{\cdot}xSiO_2{\cdot}yH_2O$ (Na_2Ze). $Na_2Ze + CaCl_2 \rightarrow CaZe + 2NaCl$; $Na_2Ze + Mg(HCO_3)_2 \rightarrow MgZe + 2NaHCO_3$. **Regeneration** (10% brine): $CaZe + 2NaCl \rightarrow Na_2Ze + CaCl_2$. Residual ~10 ppm, no sludge. **Limitations:** turbid/acidic water, Fe/Mn zeolite ko kharab karte hain; anions remove nahi hote ($NaHCO_3 \rightarrow$ caustic embrittlement).

(c) Ion Exchange (Demineralization)

Cation resin ($–SO_3H$): $2RH + Ca^{2+} \rightarrow R_2Ca + 2H^+$. **Anion resin** ($–NR_3{}^+OH^-$): $ROH + Cl^- \rightarrow RCl + OH^-$; then $H^+ + OH^- \rightarrow H_2O$. **Regeneration:** cation – dil. HCl/H_2SO_4; anion – dil. NaOH. Residual 0–2 ppm; best for high-pressure boilers; resins costly, turbid water nahi.



	Lime-soda	Zeolite	Ion exchange
Residual hardness / ions removed	15–60 ppm; Ca, Mg ppt.	~10 ppm; cations only	0–2 ppm; cations + anions
Sludge / acidic water	Yes / OK	No / not suitable	No / OK

5. Boiler Problems

	Sludge	Scale
Nature & cause	Soft, loose ppt. (colder parts); $MgCO_3$, $MgCl_2$, $CaCl_2$, $MgSO_4$	Hard adherent coating; $CaSO_4$ (solubility ↓ with temp), $CaCO_3$, $Mg(OH)_2$ [$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HCl$], $CaSiO_3$
Effects	Poor heat transfer, choking of pipes	Fuel wastage, overheating & weakening, **explosion** risk when scale cracks
Prevention	Blow-down, soft water	Softening, internal conditioning, scraping, thermal shocks, dil. HCl ($CaCO_3$) / EDTA ($CaSO_4$) wash

Caustic embrittlement: high pressure par $Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2$; NaOH cracks/rivets mein concentrate hokar iron ko **sodium ferroate** (Na_2FeO_2) ke roop mein dissolve karta hai → brittle. (Concentration cell: Fe | conc. NaOH || dil. NaOH | Fe.) **Prevention:** Na_3PO_4 instead of Na_2CO_3, tannin/lignin/Na_2SO_4 add karke cracks block.

Priming (steam ke saath water droplets – wet steam): cause high water level, high steam velocity, sudden boiling; prevention – controlled velocity & level, steam purifiers. **Foaming** (stable bubbles): cause oil, alkalis, suspended matter; prevention – anti-foam (castor oil), oil removal by $NaAlO_2$/alum, blow-down.

Boiler corrosion: (i) dissolved O_2: $4Fe + 6H_2O + 3O_2 \rightarrow 4Fe(OH)_3$; remove by $2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$, hydrazine $N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$, mechanical de-aeration; (ii) $CO_2 \rightarrow H_2CO_3$; remove by NH_4OH/lime; (iii) acids from salts: $MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HCl$; $Fe + 2HCl \rightarrow FeCl_2 + H_2$ (chain) – alkali neutralisation.

6. Conditioning of Water (Internal Treatment)

Colloidal (tannin, agar, kerosene – loose sludge); **Phosphate** (high pressure): $3CaCl_2 + 2Na_3PO_4 \rightarrow Ca_3(PO_4)_2\downarrow + 6NaCl$; **Carbonate** (low pressure): $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$; **Calgon**: $Na_2[Na_4(PO_3)_6] + 2CaSO_4 \rightarrow Na_2[Ca_2(PO_3)_6]$ (soluble) + $2Na_2SO_4$; **Sodium aluminate**: $NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3$ (flocs entrap oil/silica); electrical & radioactive conditioning; blow-down.

7. Numericals

Q1 $Ca(HCO_3)_2$ 162, $Mg(HCO_3)_2$ 73, $CaSO_4$ 136, $MgCl_2$ 95, NaCl 50 mg/L. Hardness, and lime (90% pure) & soda for 10^5 L?

$CaCO_3$ eq: 100, 50, 100, 100 (NaCl – nil). Temp = **150 ppm**, Perm = **200 ppm**, Total = **350 ppm**. Lime = $0.74[100 + 2\times50 + 100] \times 10^5 \times 100/90$ mg = **24.67 kg**; Soda = $1.06[100 + 100] \times 10^5$ mg = **21.2 kg**.

Q2 EDTA SHW 50 mL = 20 mL EDTA; sample = 15 mL; boiled = 10 mL. 1 mL EDTA = 2.5 mg $CaCO_3$. Total = 15/20×1000 = **750**, Perm = **500**, Temp = **250 ppm**.

Q3 ZEOLITE 5000 L water; regen. 20 L NaCl (100 g/L) = 2000 g ≡ 2000×50/58.5 = 1709.4 g $CaCO_3$. Hardness = $1709.4\times10^3/5000$ = **341.9 ppm**.

UNIT II Fuels and Lubricants

1. Fuels: Classification and Calorific Value

Fuel = combustible carbonaceous substance jo jalne par heat deta hai. Good fuel: high CV, moderate ignition temp, low moisture/ash, cheap, easy transport, controllable, non-polluting. **Primary:** solid – wood, peat, lignite, coal; liquid – crude oil; gas – natural gas. **Secondary:** solid – coke, charcoal; liquid – petrol, diesel, kerosene; gas – LPG, coal gas, producer/water gas, biogas.

CALORIFIC VALUE Unit mass fuel ke complete combustion par total heat (kcal/kg, kJ/kg; gas kcal/m^3). **HCV/GCV**: products room temp tak cool – steam ka latent heat included. **LCV/NCV**: water vapour escape.

$$\text{LCV} = \text{HCV} - 0.09\,\text{H} \times 587\ \text{kcal/kg} \quad | \quad \textbf{Dulong:}\ \text{HCV} = \frac{1}{100}[8080\,\text{C} + 34500\,(\text{H} - \frac{\text{O}}{8}) + 2240\,\text{S}]\ \text{kcal/kg}$$

(H − O/8) = available hydrogen. Bomb calorimeter: HCV = (W + w)(t_2 − t_1)/x.

EXAMPLE C 80%, H 5%, O 6%, S 1%: HCV = [8080×80 + 34500(5 − 0.75) + 2240×1]/100 = **7952.7 kcal/kg**; LCV = 7952.7 − 0.09×5×587 = **7688.5 kcal/kg**.

Fuel resources in India: Coal – Jharia, Bokaro (Jharkhand), Raniganj (WB), Talcher (Odisha), Korba (CG), Singrauli & Sohagpur (MP) – mostly bituminous, high ash; lignite – Neyveli (TN); petroleum – Digboi (Assam, oldest), Mumbai High, Ankleshwar (Gujarat), Barmer (Rajasthan) (~80% crude imported); natural gas – Mumbai High, KG basin; nuclear – U (Jaduguda), Th (Kerala monazite); renewables – hydro, solar, wind, biomass.

2. Coal: Proximate and Ultimate Analysis

Rank: wood (C 50%) → peat (57%) → lignite (67%) → bituminous (83%) → anthracite (93%); rank ↑ ⇒ C%, CV ↑, moisture, VM ↓.

Proximate (wt basis):
- **Moisture**: 1 g, 105–110 °C, 1 hr → loss% (CV ↓).
- **Volatile matter**: covered crucible, 925 ± 5 °C, 7 min (long smoky flame, CV ↓).
- **Ash**: open crucible, 700–750 °C (CV ↓, clinker).
- **Fixed carbon** = 100 − (M + VM + A) (FC ↑ ⇒ CV ↑).

Ultimate (elemental):
- C, H: burn in O_2; CO_2 → KOH, H_2O → $CaCl_2$.

$\%C = \Delta KOH \times 12 \times 100/(44\,W)$; $\%H = \Delta CaCl_2 \times 2 \times 100/(18\,W)$
- N (Kjeldahl): $\%N = 1.4\,V\,N\,/\,W$
- S (as $BaSO_4$): $\%S = \text{wt}\ BaSO_4 \times 32 \times 100/(233\,W)$
- O = 100 − (C + H + N + S + ash). C, H good; O, N, S bad.

3. Clinker Formation and Pulverized Coal

Clinker: ash (SiO_2, Al_2O_3, Fe_2O_3, CaO, alkalis) high temp par fuse hokar hard glassy mass ban jaati hai jo grate par chipakti hai (low ash-fusion temp ⇒ zyada clinker) → air flow block, unburnt fuel loss. Prevention: low-ash/high fusion-temp coal, temp control, steam under grate. **Pulverized coal:** coal ko ~200 mesh (< 75 µm) powder bana kar hot air blast ke saath burn karna – complete combustion with less excess air, low grade coal usable, easy control, no clinker, high efficiency (thermal power plants); demerits – grinding cost, fly ash (ESP needed), dust explosion risk.

4. Carbonization and Manufacture of Coke

Carbonization = coal ko air ki absence mein heat karna → coke + gas + tar + NH_3 (sirf caking/coking coal se strong coke). **LTC** (500–700 °C): soft semi-coke (5–15% VM, domestic), less gas but high CV (6500–9500 kcal/m^3). **HTC** (900–1200 °C): hard porous metallurgical coke (1–3% VM), more gas, lower CV (5400–6000 kcal/m^3).

Beehive oven: dome brick chamber, coal partly burns inside, by-products waste, 3–4 days, obsolete. **Otto–Hoffman by-product oven:** narrow silica chambers, **regenerative** heating (flue gas se checker bricks garam, phir air/gas preheat), 11–18 hr, water quenching; by-products recovered – tar, NH_3 [$(NH_4)_2SO_4$], naphthalene, benzene, H_2S. **Good metallurgical coke:** low ash/S/P, porous, strong, low reactivity, uniform size.

5. Petroleum, Cracking and Cracked Gasoline

Refining fractions: LPG (C_1–C_4), petrol (C_5–C_8, 40–120 °C), naphtha, kerosene (C_{10}–C_{16}, 180–250 °C), diesel (C_{15}–C_{18}, 250–320 °C), heavy oil (C_{17}–C_{30}, 320–400 °C → lube oil, wax), residue → bitumen.

Cracking = bade hydrocarbons ko heat/catalyst se chhote low-boiling molecules mein todna (more petrol): $C_{10}H_{22} \rightarrow C_5H_{12} + C_5H_{10}$.

Thermal: liquid-phase 475–530 °C, 100 kg/cm^2; vapour-phase 600–650 °C, low pressure (ON ~65–70). **Catalytic** (Al_2O_3–SiO_2/zeolite, 300–450 °C, 1–5 atm): fixed-bed or **fluid-bed (FCC)** with continuous catalyst regeneration – better yield, ON 80–100, less S and gum.

Cracked gasoline: olefins + branched alkanes + aromatics ⇒ high octane (better anti-knock than straight-run); par olefins storage mein

polymerise hokar **gum** banate hain → inhibitors. Mercaptans hatane ko **Doctor sweetening**: 2RSH + Na_2PbO_2 → $(RS)_2Pb$ + 2NaOH; $(RS)_2Pb$ + S → RSSR + PbS. Reforming (Pt) ON aur badhata hai.

Fuel oil	Range / CV	Properties	Uses
Petrol	C_5–C_8; ~11,250 kcal/kg	Volatile, high ON	SI engines
Kerosene	C_{10}–C_{16}; ~11,100	Less volatile	Domestic, jet fuel (ATF)
Diesel (HSD)	C_{15}–C_{18}; ~11,000	High cetane	CI engines
Furnace oil	C_{17}–C_{30}	Viscous, high S, preheating	Boilers, furnaces, ships
LPG	Butane + propane	Odorant ethyl mercaptan	Domestic, auto

6. Knocking, Octane and Cetane Number

Knocking (SI engine): spark ke baad bacha hua fuel-air mixture flame front aane se pehle hi **self-ignite** ho jaata hai → shock wave, metallic sound, power loss, damage. Tendency: n-paraffins > branched paraffins > olefins > cycloparaffins > aromatics. **Octane number** = % iso-octane (2,2,4-trimethylpentane, ON 100) in iso-octane + n-heptane (ON 0) mixture jo same knocking de. **Diesel knock** (CI): ignition delay se fuel jama hokar ek saath jalta hai; straight-chain best. **Cetane number** = % n-cetane ($C_{16}H_{34}$, CN 100) in cetane + α-methylnaphthalene (CN 0) mixture. High ON ⇒ low CN.

Anti-knock: TEL $(C_2H_5)_4Pb$ + ethylene dibromide (Pb → volatile $PbBr_2$); toxic, catalytic converter poison → now unleaded with MTBE, ethanol (E10/E20), aromatics. **Cetane improvers:** ethyl/amyl nitrate, acetone peroxide.

7. Combustion Calculations

FORMULA $C + O_2 \rightarrow CO_2$ (12:32); $2H_2 + O_2 \rightarrow 2H_2O$ (4:32); $S + O_2 \rightarrow SO_2$ (32:32)

$$\text{Min } O_2\ (\text{kg/kg}) = \frac{32}{12}C + 8\left(H - \frac{O}{8}\right) + S\ ;\quad \text{Air (wt)} = O_2 \times \frac{100}{23}\ ;\quad \text{Air (vol)} = O_2\ \text{vol} \times \frac{100}{21}$$

O_2 vol per vol gas: H_2 ½, CO ½, CH_4 2, C_2H_2 2.5, C_2H_4 3, C_2H_6 3.5, C_3H_8 5. 1 kmol = 22.4 m^3 (NTP). Actual air = min × (1 + excess).

Q1 Coal C 80, H 5, O 6, S 1%. O_2 = 2.667×0.80 + 8(0.05 − 0.0075) + 0.01 = 2.483 kg; Air = 2.483×100/23 = **10.80 kg/kg**.

Q2 Gas H_2 50, CH_4 30, CO 10, N_2 6, CO_2 4%. O_2 = 0.25 + 0.60 + 0.05 = 0.90 m^3; air = 4.286 m^3; with 20% excess = **5.14 m^3**.

8. Lubricants

Lubricant = substance jo moving surfaces ke beech friction & wear kam kare. Functions: friction/wear ↓, cooling, corrosion protection, sealing, power loss ↓.

Mechanism	Condition	Working	Lubricant
Thick film (hydrodynamic)	Low load, high speed	≥ 1000 Å oil film; surfaces touch nahi; viscosity decides	Mineral oils (turbines, watches)
Thin film (boundary)	High load, low speed	Thin adsorbed layer; oiliness important	Veg./animal oils, graphite, MoS_2
Extreme pressure	Very high load/temp	Additives form $FeCl_2$, FeS, phosphide films	Chlorinated esters, sulphurised oils, tricresyl phosphate

Classification: (1) **Liquid** – animal/vegetable oils (good oiliness, oxidise), mineral oils (cheap, stable), blended oils (+ additives), synthetic (silicones). (2) **Semi-solid (greases)** – oil + metal soap: Ca-soap (water resistant, < 80 °C), Na-soap (high temp), Li-soap (multipurpose); for heavy load, low speed, rail axles. (3) **Solid** – graphite (layers, oildag/aquadag), MoS_2, talc, mica; very high temp/load.

Properties & testing: viscosity (Redwood No.1/2, Saybolt, Engler viscometers); **viscosity index** VI = (L − U)/(L − H) × 100 (high = good); **flash & fire point** (Pensky-Martens closed cup, Cleveland open cup); **cloud & pour point** (low for cold); **oiliness**; **aniline point** (high = less aromatics, seals safe); **carbon residue** (Conradson); **neutralization number** (mg KOH/g, < 0.1); **saponification value**; **steam emulsion number**; volatility, four-ball EP test.

Applications: watches/instruments – light oils; IC engines – multigrade oils + additives; turbines – high VI, low emulsion; gears – EP lubricants; bearings/axles – greases; furnaces, vacuum, space – graphite/MoS_2; machining – cutting-fluid emulsions.

UNIT III Free Energy, Equilibrium, Batteries and Corrosion

1. Thermodynamic Functions and Free Energy

Internal energy U (state function): ΔU = q + w (w = −PΔV). **Enthalpy** H = U + PV, $\Delta H = q_p = \Delta U + \Delta n_g RT$. **Entropy** S = randomness, $\Delta S = q_{rev}/T$; spontaneous process mein $\Delta S_{total} > 0$. **Gibbs free energy** G = H − TS, **ΔG = ΔH − TΔS** = max useful work ($-\Delta G = w_{max}$).

Helmholtz A = U − TS.

ΔH	ΔS	Spontaneous?	ΔH	ΔS	Spontaneous?
−	+	Always	−	−	Low T only
+	−	Never	+	+	High T only

FREE ENERGY & EQUILIBRIUM

$$\Delta G = \Delta G^\circ + RT \ln Q \text{ ; at equilibrium } \Delta G = 0 \Rightarrow \Delta G^\circ = -2.303\ RT \log K \text{ ; } \Delta G = -nFE, \Delta G^\circ = -nFE^\circ$$

ΔG < 0 spontaneous, = 0 equilibrium, > 0 non-spontaneous; ΔG° < 0 ⇒ K > 1. F = 96500 C/mol.

2. EMF, Cell Potential and Nernst Equation

Galvanic cell chemical → electrical energy: anode (oxidation, −), cathode (reduction, +), salt bridge. **EMF** = zero current par electrodes ka potential difference. Reference SHE (E° = 0). Daniell: Zn | Zn^{2+} || Cu^{2+} | Cu, $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ = 0.34 − (−0.76) = **1.10 V**.

NERNST EQUATION

$$E = E^\circ - \frac{RT}{nF} \ln Q = E^\circ - \frac{0.0591}{n} \log Q \ (298\ K) \text{ ; electrode: } E = E^\circ + \frac{0.0591}{n} \log [M^{n+}] \text{ ; } \log K = \frac{nE^\circ}{0.0591}$$

Applications: electrode potential at any conc.; **pH** (H-electrode E = −0.0591 pH, glass electrode); equilibrium constant; **concentration cell** E = (0.0591/n) log (C_2/C_1); K_{sp}; potentiometric titrations; corrosion prediction.

EXAMPLE Daniell, $[Zn^{2+}]$ = 0.1, $[Cu^{2+}]$ = 0.01 M: E = 1.10 − (0.0591/2) log 10 = **1.070 V**. ΔG° = −2×96500×1.10 = **−212.3 kJ**; log K = 2.20/0.0591 = 37.2 ⇒ K ≈ **1.7×10^{37}**.

3. Batteries: Primary, Secondary, Thermal, Fuel Cell

Types: primary (non-rechargeable – dry cell, alkaline, Li primary), **secondary** (rechargeable – lead-acid, Ni-Cd, Li-ion), **reserve** (activated when needed – thermal battery), **fuel cells** (continuous fuel supply).

Cell	Anode / Cathode / Electrolyte	Reactions	V
Dry (Leclanché)	Zn / MnO_2+C / NH_4Cl+$ZnCl_2$	$Zn \rightarrow Zn^{2+} + 2e^-$; $2MnO_2 + 2NH_4^+ + 2e^- \rightarrow Mn_2O_3 + 2NH_3 + H_2O$	1.5
Alkaline	Zn / MnO_2 / KOH	$Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$; $2MnO_2 + H_2O + 2e^- \rightarrow Mn_2O_3 + 2OH^-$	1.5
Lead–acid	Pb / PbO_2 / H_2SO_4 (sp. gr. 1.28)	$Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$; $PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$. Overall $Pb + PbO_2 + 2H_2SO_4 \rightleftharpoons 2PbSO_4 + 2H_2O$ (sp. gr. falls on discharge)	2.0
Ni–Cd	Cd / NiO(OH) / KOH	$Cd + 2OH^- \rightarrow Cd(OH)_2 + 2e^-$; $2NiO(OH) + 2H_2O + 2e^- \rightarrow 2Ni(OH)_2 + 2OH^-$	1.2–1.4
Li-ion	LiC_6 / $LiCoO_2$ / $LiPF_6$ (organic)	$LiC_6 \rightarrow C_6 + Li^+ + e^-$; $CoO_2 + Li^+ + e^- \rightarrow LiCoO_2$ (rocking chair)	3.7

Thermal battery (reserve primary): solid electrolyte LiCl–KCl eutectic room temp par non-conducting (shelf life 10–20 yr). **Pyrotechnic heat source** (Fe + $KClO_4$) ignite hokar electrolyte ko ~400–500 °C par melt karta hai. Anode Li-alloy (Li–Si), cathode FeS_2, ~2 V/cell, high power, short life. Uses: missiles, military, aircraft emergency, space.

H_2–O_2 fuel cell: porous C electrodes (Pt/Ni), KOH electrolyte. A: $2H_2 + 4OH^- \rightarrow 4H_2O + 4e^-$; C: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; overall $2H_2 + O_2 \rightarrow 2H_2O$, **1.23 V**. 60–70% efficient, pollution free (Apollo); costly Pt, H_2 storage. Types: AFC, PEMFC, PAFC, MCFC, SOFC, methanol FC.

4. Corrosion: Theories and Mechanism

Corrosion = metal ka environment se chemical/electrochemical reaction se gradual destruction (rusting $Fe_2O_3 \cdot xH_2O$) – reverse of metallurgy.

(a) Dry (chemical) corrosion – gases se direct attack, high temp. Oxidation: $2M \rightarrow 2M^{n+} + 2ne^-$; $nO_2 + 4ne^- \rightarrow 2nO^{2-}$. Oxide film: stable non-porous (Al, Cr) – protective; unstable (Ag, Au, Pt) – no corrosion; volatile (MoO_3) & porous – continuous. **Pilling–Bedworth ratio** = vol. oxide / vol. metal: < 1 porous, non-protective (Li, Na, K, Mg); ≥ 1 protective (Al, Cr). Also Cl_2, H_2S, SO_2 attack, liquid metal corrosion.

(b) Wet (electrochemical) corrosion – electrolyte ki presence mein anodic & cathodic areas. Anode: $Fe \rightarrow Fe^{2+} + 2e^-$. Cathode: **$H_2$ evolution** (acidic, no O_2): $2H^+ + 2e^- \rightarrow H_2$; **$O_2$ absorption** (neutral, aerated): $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$; $Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$; $4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$ → rust. **Types:** galvanic (dissimilar metals – active one anode), **differential aeration** (kam O_2 wala part anode – waterline, under dirt), pitting, crevice, stress corrosion, intergranular.

Dry	No moisture; direct chemical; uniform; slow; product at attack site
Wet	Electrolyte; electrochemical cells; only anode corrodes; fast; product between anode & cathode

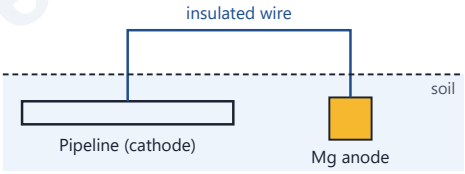
5. Factors Affecting Rate of Corrosion

Metal: position in galvanic series (Mg, Zn, Al, Cd, steel, Pb, Sn, brass, Cu, Ni, Ag, Ti, Au, Pt – active → noble; zyada difference = fast); **small anode + large cathode** = intense; impurity, stressed parts, nature of oxide film, solubility/volatility of products, over-voltage.

Environment: temperature ↑, humidity (critical ~60–80%), low pH, impurities (SO_2, H_2S, Cl^- near sea), conductance, O_2 concentration difference.

6. Protection: Cathodic, Anodic and Other Measures

Cathodic protection – structure ko cathode banana. (i) **Sacrificial anode:** Mg/Zn/Al block jodo; woh corrode hota hai, structure bachta hai (pipelines, ship hulls, tanks). (ii) **Impressed current:** DC source ka −ve structure se, +ve inert anode (graphite, scrap iron in gypsum backfill) se (long pipelines, tanks).
Anodic protection: potentiostat se anodic current dekar metal ko **passive region** mein rakhna (protective oxide). Only active–passive metals – stainless steel, Ti, steel in H_2SO_4 tanks.



Sacrificial anode protection

Other measures: proper design (avoid dissimilar metal contact, large anode/small cathode, no crevices, drainage); material selection (pure metals, stainless steel Cr ≥ 12%); modifying environment – deaeration, dehumidification, **inhibitors**: anodic (chromates, phosphates), cathodic (amines, As/Sb salts), vapour-phase (dicyclohexylammonium nitrite); protective coatings.

7. Protective Coatings

Galvanizing (anodic, Zn on Fe)	Tinning (cathodic, Sn on Fe)
Pickle (dil. H_2SO_4) → NH_4Cl flux → molten Zn 425–450 °C → rollers	Pickle → $ZnCl_2$ flux → molten Sn → palm oil
Scratch par bhi Zn sacrificially protect karta hai	Scratch par small Fe anode → rapid corrosion
Not for food (Zn toxic); pipes, roofing, buckets	Non-toxic; food cans, utensils

Other metallic: electroplating (Ni, Cr, Cu, Au; object = cathode), cladding (alclad), metal spraying, cementation – sherardizing (Zn), chromizing (Cr), calorizing (Al). **Inorganic non-metallic:** phosphating (H_3PO_4 + Zn/Mn phosphate – car body paint base), chromating, **anodizing** (Al anode in H_2SO_4/chromic acid → thick Al_2O_3), vitreous enamel, cement coatings. **Organic: paint** = pigment (ZnO, TiO_2, red lead) + vehicle/drying oil (linseed – oxidation & polymerisation) + thinner (turpentine) + drier (Co/Mn naphthenates) + extender ($BaSO_4$) + plasticizer + anti-skinning agent; varnish (resin in oil/solvent, no pigment), lacquer, enamel (pigment + varnish), polymer/rubber, bitumen coatings.

UNIT IV Cement, Refractories and High Polymers

1. Cement: Types and Manufacture of Portland Cement

Types: natural, puzzolana (PPC – fly ash), slag, **OPC** (33/43/53 grade), rapid hardening (more C_3S), low heat (dams), sulphate resisting (C_3A < 5%), white (Fe-free), high alumina, quick setting. **Portland cement composition:** CaO 60–67, SiO_2 17–25, Al_2O_3 3–8, Fe_2O_3 0.5–6, MgO 0.1–4, SO_3 1–3%.

Manufacture: calcareous (limestone) + argillaceous (clay) → mixing by **dry** (powder, less fuel) or **wet** process (slurry, better mixing) → **rotary kiln** (inclined rotating steel cylinder, flame at lower end) → clinker → cooled, ground with 2–3% **gypsum**. Kiln zones: **drying** (~400 °C, water out) → **calcination** (700–1000 °C, $CaCO_3 \rightarrow CaO + CO_2$) → **clinkering** (1400–1500 °C: $2CaO + SiO_2 \rightarrow C_2S$; $3CaO + SiO_2 \rightarrow C_3S$; $3CaO + Al_2O_3 \rightarrow C_3A$; $4CaO + Al_2O_3 + Fe_2O_3 \rightarrow C_4AF$).

Compound	C_3S $3CaO·SiO_2$ (~45%)	C_2S $2CaO·SiO_2$ (~25%)	C_3A $3CaO·Al_2O_3$ (~10%)	C_4AF $4CaO·Al_2O_3·Fe_2O_3$ (~9%)
Role	Early strength (1–7 d)	Later strength (7–28 d)	Initial/flash set, max heat	Early set, grey colour

2. Setting and Hardening of Cement

REACTIONS **Setting** = plasticity loss; **hardening** = strength gain (hydration + gel formation + crystallisation).
Initial set (1 day): $3CaO·Al_2O_3 + 6H_2O \rightarrow 3CaO·Al_2O_3·6H_2O$ + 880 kJ/kg; $C_4AF + 7H_2O \rightarrow 3CaO·Al_2O_3·6H_2O + CaO·Fe_2O_3·H_2O$
Gypsum (retarder): $3CaO·Al_2O_3 + 3CaSO_4·2H_2O + 26H_2O \rightarrow 3CaO·Al_2O_3·3CaSO_4·32H_2O$ (**ettringite**, stops flash set)
Hardening (1–7 d): $2(3CaO·SiO_2) + 6H_2O \rightarrow 3CaO·2SiO_2·3H_2O$ (**tobermorite gel**) + $3Ca(OH)_2$
Hardening (7–28 d): $2(2CaO·SiO_2) + 4H_2O \rightarrow 3CaO·2SiO_2·3H_2O + Ca(OH)_2$

Properties of OPC: initial setting ≥ 30 min, final ≤ 600 min; soundness (Le-Chatelier ≤ 10 mm); fineness ≥ 225 m^2/kg; 28-day strength 33/43/53 MPa; heat of hydration. **Uses:** concrete, RCC, mortar, plaster, dams, bridges, roads, pipes.

3. Refractories

Refractory = inorganic material jo high temperature par bina soften/deform hue load, abrasion aur slag attack sah sake – furnace, kiln, converter linings. **Classification: acidic** – silica, fireclay (acid slag resist; for coke ovens, glass furnace roofs); **basic** – magnesite (MgO), dolomite (basic slag; open hearth, LD converter); **neutral** – chromite, graphite, SiC, zirconia (both; separating layer, crucibles). By refractoriness: normal 1580–1780 °C, high 1780–2000 °C, super > 2000 °C.

Properties: refractoriness (softening temp – **Seger cone/PCE** test), refractoriness under load (RUL, 3.5 kg/cm^2), porosity (low = strong, slag resistant), **thermal spalling** resistance, dimensional stability, suitable thermal conductivity, low expansion, chemical inertness, crushing strength. **Uses:** blast furnace, coke ovens, cement/glass kilns, boilers, crucibles, rocket nozzles, nuclear reactors.

4. High Polymers: Classification and Mechanism

Polymer = monomers ke covalent bonding se bani macromolecule; **degree of polymerization** = no. of repeating units; **functionality** = reactive sites (2 → linear, 3 → cross-linked). **Classification:** origin – natural (rubber, cellulose), semi-synthetic, synthetic; structure – linear (HDPE), branched (LDPE), cross-linked (bakelite); mode – addition vs condensation; homo- vs co-polymer; thermoplastic vs thermosetting; tacticity – iso-, syndio-, atactic.

Free radical (chain) addition (initiator benzoyl peroxide):
Initiation: $(C_6H_5COO)_2 \rightarrow 2C_6H_5\cdot + 2CO_2$; $R\cdot + CH_2{=}CH_2 \rightarrow RCH_2CH_2\cdot$
Propagation: $RCH_2CH_2\cdot + nCH_2{=}CH_2 \rightarrow R(CH_2CH_2)_nCH_2CH_2\cdot$
Termination: combination or disproportionation.

Ionic: cationic (BF_3/H_2O – isobutylene), anionic (n-BuLi, KNH_2 – acrylonitrile). **Coordination:** Ziegler–Natta $TiCl_4 + Al(C_2H_5)_3$ → linear, stereoregular. **Condensation (step-growth):** bifunctional monomers, small molecule (H_2O, HCl) eliminated – nylon, bakelite.

5. PE, PVC, PMMA, Nylons and Bakelite

Polymer	Preparation	Properties	Uses
PE – $(CH_2CH_2)_n$–	LDPE: 1000–3000 atm, 200–250 °C, O_2/peroxide (branched). HDPE: Ziegler–Natta, 6–7 atm, 60–70 °C (linear)	Inert, insulator, tough; HDPE rigid, denser (0.94–0.97)	Bags, films; buckets, pipes, containers
PVC – $(CH_2CHCl)_n$–	Vinyl chloride + benzoyl peroxide (free radical)	Rigid, flame/chemical resistant; plasticized (DOP) flexible	Pipes, cable insulation, floor tiles, raincoats
PMMA (Perspex)	$CH_2{=}C(CH_3)COOCH_3$, free radical (acetyl peroxide)	Highly transparent, light, weather resistant	Lenses, aircraft windows, signboards
Nylon-6,6	$nH_2N(CH_2)_6NH_2 + nHOOC(CH_2)_4COOH \rightarrow [–NH(CH_2)_6NHCO(CH_2)_4CO–]_n + 2nH_2O$	Polyamide, H-bonds → strong, tough, m.p. ~265 °C	Fibres, ropes, tyre cord, gears
Nylon-6	Caprolactam ring opening (~260 °C) → $[–NH(CH_2)_5CO–]_n$	m.p. ~220 °C	Tyre cord, fabrics
Bakelite	Phenol + HCHO → o/p-methylol phenol → novolac (linear) → + hexamethylenetetramine, heat → cross-linked	Thermoset, hard, heat & electrical insulator	Switches, plugs, handles, laminates

6. Plastics: Preparation, Engineering Properties and Uses

Thermoplastics (PE, PVC, PMMA, nylon): linear, heat se soft – reshape & recycle possible. **Thermosets** (bakelite, epoxy, UF): 3-D cross-linked, heating par permanently hard, insoluble, not recyclable.
Preparation (compounding & moulding): resin (binder) + plasticizer (DOP) + filler (wood flour, mica) + pigment + stabilizer (Pb salts for PVC) + catalyst/accelerator + lubricant; moulded by compression (thermosets), injection (thermoplastics), transfer, extrusion (pipes), blow moulding (bottles), calendering (sheets).
Engineering properties: low density, high strength/weight, corrosion & chemical resistance, electrical/thermal insulation, easy fabrication, transparency, low cost; limitations – low heat resistance, creep, flammable, UV degradation, non-biodegradable. **Uses:** auto & aircraft parts, electrical fittings, pipes, packaging, medical devices.

7. Natural and Synthetic Rubbers; Compounding and Vulcanization

Natural rubber = **cis-1,4-polyisoprene** from latex of *Hevea brasiliensis*, coagulated by acetic/formic acid (gutta-percha = trans form). Raw rubber drawbacks: sticky in summer, brittle in winter, low tensile strength, swells in solvents, oxidises, high water absorption.
Compounding: vulcanizing agent (S), accelerators (MBT, TMTD) + activator (ZnO), antioxidants, reinforcing filler (**carbon black**), softeners/plasticizers, colours. **Vulcanization** (Goodyear 1839): rubber + 3–5% S at 100–140 °C → allylic positions par **–S–S– cross-links** → tensile strength ~10×, elasticity ↑, non-sticky, wide temp range (−40 to 100 °C), solvent & abrasion resistance ↑. 30–50% S → **ebonite** (hard).

Synthetic rubber	Monomers	Property → Uses
Buna-S (SBR)	Butadiene 75% + styrene 25%	Abrasion resistant → tyres, soles
Buna-N (NBR)	Butadiene + acrylonitrile	Oil resistant → seals, fuel hoses
Neoprene	Chloroprene (2-chloro-1,3-butadiene)	Oil, flame, weather resistant → belts, cables
Butyl	Isobutylene ~98% + isoprene ~2%	Gas impermeable → inner tubes
Thiokol	Ethylene dichloride + Na-polysulphide	Solvent resistant → hoses, rocket fuel binder

8. Silicones and Adhesives

Silicones = $(R_2SiO)_n$, backbone **–Si–O–Si–** with organic groups on Si. Rochow: $2CH_3Cl + Si \rightarrow (CH_3)_2SiCl_2$ (Cu, 300 °C); hydrolysis → $(CH_3)_2Si(OH)_2 + 2HCl$; condensation → $–[Si(CH_3)_2O]_n–$. R_3SiCl chain stopper, $RSiCl_3$ cross-linker. **Properties:** thermally stable (−90 to 250 °C), water repellent, chemically & physiologically inert, good insulator. **Uses:** oils (lubricant, hydraulic fluid), greases, rubbers (gaskets, implants), resins (insulating varnish, water-proofing).

Adhesives = substances jo surfaces ko surface attachment se jodte hain; adhesion by mechanical interlocking, adsorption, chemical bonding, diffusion. **Natural:** starch, animal glue, casein, shellac. **Synthetic: epoxy** (bisphenol-A + epichlorohydrin, amine hardener – metal, glass), phenol/urea-formaldehyde (plywood), PVA (wood), cyanoacrylate (super glue), polyurethane, neoprene-based. Factors: clean rough surface, wetting, curing. Uses: plywood, packaging, aircraft/auto assembly, electronics, footwear.

UNIT V Environmental Chemistry and Instrumental Techniques

A1. Pollution and Air Pollution

Pollution = air/water/soil mein undesirable change jo living beings ke liye harmful ho. Pollutants: primary (SO_2, CO) & secondary (O_3, PAN, H_2SO_4). **Causes:** population, industrialisation, vehicles, fossil fuels, deforestation, chemicals, poor waste disposal.

Air pollutants: CO (incomplete combustion → carboxyhaemoglobin, O_2 transport block); **SO_x** (coal → respiratory, acid rain); **NO_x** (vehicles → smog, acid rain); **hydrocarbons** (carcinogenic); **PM_{10}/$PM_{2.5}$** (lungs, asthma); Pb, CFCs. **Types:** natural vs anthropogenic, indoor vs outdoor, gaseous vs particulate. **Classical smog** (smoke + fog + SO_2, reducing). **Photochemical smog** (oxidizing): $NO_2 + h\nu \rightarrow NO + O$; $O + O_2 \rightarrow O_3$; $O_3 + HC \rightarrow HCHO$, **PAN** (eye irritation). **Acid rain** (pH < 5.6): $2SO_2 + O_2 + 2H_2O \rightarrow 2H_2SO_4$; $4NO_2 + O_2 + 2H_2O \rightarrow 4HNO_3$ – marble (Taj Mahal) damage.

A2. Green House Effect and Ozone Layer

Greenhouse effect: sun ki short-wave radiation earth ko garam karti hai; earth **IR** re-emit karti hai jise CO_2, CH_4, N_2O, CFCs, O_3, H_2O vapour absorb karke wapas bhejte hain. Natural effect earth ko ~15 °C rakhta hai (warna −18 °C); excess GHG → **global warming** – ice melting, sea level rise, extreme weather, crop loss. Control: renewable energy, afforestation, efficiency, Kyoto (1997), Paris (2015).

Ozone layer (stratosphere ~15–35 km, ~300 DU) harmful **UV-B/UV-C** (200–315 nm) absorb karti hai → skin cancer, cataract, immune & crop damage se bachav. Formation: $O_2 + h\nu$ ($\lambda < 242$ nm) → 2O; $O + O_2 + M \rightarrow O_3 + M$. **Depletion by CFCs:** $CF_2Cl_2 + h\nu \rightarrow \cdot CF_2Cl + Cl\cdot$; $Cl\cdot + O_3 \rightarrow ClO\cdot + O_2$; $ClO\cdot + O \rightarrow Cl\cdot + O_2$ (one Cl destroys ~10^5 O_3); $NO + O_3 \rightarrow NO_2 + O_2$. Antarctic ozone hole; **Montreal Protocol (1987)**.

A3. Control of Air Pollution

Particulates: settling chamber, **cyclone separator**, bag filter, wet scrubber, **electrostatic precipitator** (particles charged by corona discharge, collected on plates, > 99% efficient). **Gases:** absorption ($SO_2 + CaCO_3$ lime slurry scrubbing), adsorption (activated carbon), incineration, **catalytic converter** (Pt–Pd–Rh: $2CO + O_2 \rightarrow 2CO_2$; $2NO + 2CO \rightarrow N_2 + 2CO_2$). **Others:** CNG/LPG, low-S fuel, BS-VI, tall chimneys, green belts, Air Act 1981.

A4. Water Pollution: Sources and Prevention

Sources: domestic sewage (organics, pathogens, detergents); industrial effluents – heavy metals (Hg → Minamata, Cd → itai-itai, As, Pb), dyes, acids; agricultural runoff – fertilizers → **eutrophication**, pesticides (DDT biomagnification); thermal pollution; oil spills; radioactive waste. Indicators: **BOD** (O_2 used by microbes, 5 days, 20 °C; clean < 5 ppm), **COD** ($K_2Cr_2O_7$ oxidation; COD > BOD), DO (> 4–6 ppm good).

Prevention/treatment: primary (screening, grit, sedimentation) → secondary (biological: activated sludge, trickling filter, oxidation pond) → tertiary (chemical precipitation, ion exchange, RO, disinfection); ETPs in industries, controlled fertilizer use, no dumping in rivers, Water Act 1974.

A5. Soil/Land and Radioactive Pollution

Soil pollution causes: solid & plastic waste, industrial/hazardous & e-waste, excess fertilizers & pesticides, mining, acid rain, sewage sludge, deforestation (erosion), salinity. **Control:** 3R (reduce, reuse, recycle), waste segregation, composting/vermicomposting, sanitary landfill, incineration, bio-fertilizers/bio-pesticides, organic farming, afforestation, bioremediation.

Radioactive pollution sources: natural (cosmic rays, radon, ^{40}K, U/Th rocks); man-made (nuclear power & waste, weapon tests, U mining, medical X-rays, accidents – Chernobyl 1986, Fukushima 2011). **Effects:** somatic (burns, leukaemia, cancer) & genetic (mutations); ^{90}Sr → bones, ^{131}I → thyroid, ^{137}Cs → muscles. **Control:** safe disposal (vitrification + deep geological storage), shielding (Pb, concrete), time–distance–shielding, leak-proof reactors, monitoring (film badges/dosimeters), dose limits (AERB/ICRP).

B1. Spectroscopy and Lambert–Beer's Law

Spectroscopy = matter aur EM radiation ke interaction (absorption/emission) ka study. $E = h\nu = hc/\lambda = hc\bar{\nu}$. Regions: UV 200–400 nm, visible 400–800 nm (electronic transitions); IR 4000–400 cm^{-1} (vibrations); microwave (rotations); radio (nuclear spin – NMR).

LAMBERT–BEER'S LAW Monochromatic light ki absorbance path length (Lambert) aur concentration (Beer) ke proportional:

$$A = \log \frac{I_0}{I} = \varepsilon\, c\, l\,; \quad T = I/I_0\,; \quad A = 2 - \log(\%T)$$

ε = molar absorptivity (L mol^{-1} cm^{-1}). Valid only for dilute solutions (< 0.01 M), monochromatic light, no association/dissociation, scattering or fluorescence. **Ex:** A = 0.50, ε = 5000, l = 1 cm ⇒ c = **1×10^{-4} M**; %T = 10 ⇒ A = **1**.

B2. UV–Visible Spectroscopy

Principle: UV/visible absorb karke valence electrons bonding/non-bonding se antibonding orbital mein jaate hain; energy order **σ→σ* > n→σ* > π→π* > n→π***. **Chromophore** (C=C, C=O, NO_2, N=N) – absorption ka group; **auxochrome** (–OH, $–NH_2$) – λ_{max}/intensity badhata hai. Shifts: bathochromic (red), hypsochromic (blue), hyperchromic, hypochromic. Conjugation ↑ ⇒ λ_{max} ↑ (butadiene 217 nm). **Instrument:** H_2/D_2 lamp (UV), tungsten (visible) → monochromator → **quartz** cell → photomultiplier → recorder. **Applications:** conjugation/chromophore detection, quantitative analysis, impurity detection (benzene in alcohol), cis-trans isomers, kinetics.

B3. IR Spectroscopy

Principle: IR (4000–400 cm^{-1}) absorb hone par bonds vibrate karte hain; sirf woh vibration IR active jisme **dipole moment change** ho (N_2, O_2, H_2 inactive). Modes: stretching (sym., asym.), bending (scissoring, rocking, wagging, twisting); **3N − 6** (non-linear), **3N − 5** (linear): H_2O 3, CO_2 4. Hooke's law $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$. Functional group region 4000–1500, **fingerprint** 1500–400 cm^{-1}. Instrument: Nernst glower/globar → NaCl/KBr optics → thermocouple/bolometer; KBr pellet sample.

Bond	O–H	N–H	C–H	C≡N	C≡C	C=O	C=C	C–O
cm^{-1}	3200–3600	3300–3500	2850–3000	~2250	2100–2260	1650–1750	1620–1680	1000–1300

Applications: functional groups, compound identity (fingerprint), purity, H-bonding, reaction progress, polymer & pollutant analysis.

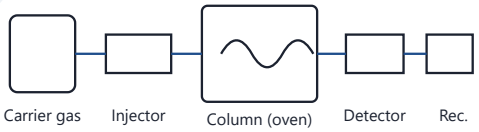
B4. NMR Spectroscopy

Principle: spin I ≠ 0 nuclei (^1H, ^{13}C; I = ½) external field B_0 mein do energy states (α aligned, β opposed) mein split hote hain; RF absorb karke spin flip – **resonance** $\nu = \gamma B_0/2\pi$. Electron cloud **shielding** → upfield; electronegative groups **deshielding** → downfield. **Chemical shift** $\delta = (\nu_{sample} - \nu_{TMS})/\nu_{spectrometer} \times 10^6$ ppm; reference **TMS** $Si(CH_3)_4$, δ = 0 (12 equivalent, highly shielded H; inert, volatile). δ: CH_3 0.9, CH–C=O 2–2.5, OCH_3 3.3–4, aromatic 6.5–8, CHO 9–10, COOH 10–12. **Signals** = types of equivalent H; **area** ∝ no. of H; **splitting** (n + 1) rule. Ethanol: CH_3 triplet, CH_2 quartet, OH singlet (3:2:1). Instrument: magnet, RF transmitter, sweep coil, receiver; solvent $CDCl_3/CCl_4$. **Applications:** structure elucidation, isomers, purity, tautomerism, **MRI**.

B5. Chromatography and Gas Chromatography

Chromatography (Tswett): mixture components ko **stationary** aur **mobile phase** ke beech different adsorption/partition ke basis par separate karna. Types: column, paper, TLC, GC, HPLC, ion-exchange. $\mathbf{R_f}$ = distance by solute / distance by solvent front. Uses: separation, purification, identification (amino acids, dyes, drugs, forensics).

Gas chromatography: mobile phase = inert carrier gas (He, N_2, Ar); stationary = solid (**GSC**) or liquid on support (**GLC**, partition). Carrier gas → heated injector → column in thermostated oven → detector (**TCD, FID, ECD**) → recorder → chromatogram. **Retention time** → identity; **peak area** → amount. Applications: petroleum, essential oils, air pollutants, pesticide residues, blood alcohol, drugs. Only volatile, thermally stable samples.



Gas chromatograph

B6. Colorimetry

Coloured solution ki concentration uski **visible absorbance** se (Beer–Lambert). **Colorimeter:** tungsten lamp → **filter** (complementary colour) → cuvette → photocell → galvanometer. Standards se **calibration curve** (A vs c, straight line), unknown ki A se c. Colourless ions pehle coloured complex mein. **Applications:** Fe (thiocyanate/phenanthroline), Cu (NH_3 blue), NH_3 (Nessler's), nitrite, phosphate, Mn (as MnO_4^-), glucose/haemoglobin in blood.

YAAD RAKHO UV-Vis = electronic (conjugation); IR = vibrations (functional groups); NMR = nuclear spin (H-environment, δ vs TMS); Colorimetry = Beer–Lambert in visible; GC = volatile mixtures (t_R + peak area).